



ENSAE - 3rd year, Master 2
Exam 2023-2024

RISK MEASURES

Closed book examination, no calculator - 2 hours

The exercises are independent. Each question counts for approximately 2 points. You can answer in English or in French.

Exercise 1 (General questions about the course)

To answer the following questions, recall the relevant definitions and/or give illustrative examples.

1. In the relation $\{x : F(x) \geq \alpha\} = [F^-(\alpha), +\infty)$, what do $F(\cdot)$ and $F^-(\cdot)$ mean?
2. Is it true that $F(x) < \alpha \Leftrightarrow x < F^-(\alpha)$?
3. Is it true that $x > F^-(\alpha) \Rightarrow F(x) > \alpha$? Hint: consider for example the cdf F of a Bernoulli variable with parameter $1 - \alpha$ and compute $F^-(\alpha)$.
4. At the risk level $\alpha \in (0, 1)$, the Expected Shortfall of a $\mathcal{N}(0, 1)$ Gaussian distribution is $\frac{1}{\alpha}\phi(\Phi^{-1}(1 - \alpha))$, with the usual notation. What are the Value-at-Risk and Expected Shortfall of a $\mathcal{N}(m, \sigma^2)$ distribution?
5. Let X and Y be two real random variables such that $Y = e^X$. What is the relationship between the quantiles $F_X^-(\alpha)$ and $F_Y^-(\alpha)$ of X and Y ?
6. Explain why a standard deviation is not considered as risk measure.
7. Can a spherical distribution have fat tails? For a bivariate elliptic distribution with cdf F , what are the shapes of the sets $F^-(\alpha)$?
8. Consider an iid sequence of real random losses $L_1, \dots, L_n$. Without looking at these losses, for $t = 1, \dots, n$ we randomly set $\text{VaR}_t = +\infty$ with probability $1 - \alpha$ and $\text{VaR}_t = -\infty$ with probability α . Is this reasonable? Do standard backtests reject this way of evaluating a VaR?

Exercise 2 (Value-at-Risk for heavy-tailed distributions)

A function $F : \mathbb{R} \rightarrow \mathbb{R}$ is said to be slowly varying (at $+\infty$), whenever for all $t > 0$

$$\lim_{x \rightarrow \infty} \frac{L(tx)}{L(x)} = 1.$$

Let X_1 and X_2 be two positive, independent and identically distributed random variables such that, for $\gamma > 0$,

$$P(X_i > x) \sim x^{-\gamma} L(x), \quad x \rightarrow \infty.$$

1. Show that

$$P(X_1 + X_2 > x) \sim 2x^{-\gamma} L(x), \quad x \rightarrow \infty.$$

Hint: we can use $\{X_1 + X_2 > x\} \subset \{X_1 > (1 - \delta)x\} \cup \{X_2 > (1 - \delta)x\} \cup \{X_1 > \delta x, X_2 > \delta x\}$ for all $\delta \in (0, 1/2)$.

Since X_1 and X_2 are positive, $\{X_1 + X_2 > x\} \supset \{X_1 > x\} \cup \{X_2 > x\}$ and thus $P(X_1 + X_2 > x) \geq 2P(X_1 > x)$. Now, using the indication,

$$P(X_1 + X_2 > x) \leq 2 \frac{1}{\{(1 - \delta)x\}^\gamma} L((1 - \delta)x) + \frac{1}{\delta^{2\gamma} x^{2\gamma}} L^2(\delta x).$$

We thus have

$$\lim_{x \rightarrow \infty} \frac{P(X_1 + X_2 > x)}{P(X_1 > x)} \leq 2 \frac{1}{(1 - \delta)^\gamma}.$$

The right hand side is arbitrarily close to 2 when $\delta > 0$ is chosen small enough. The result follows.

2. Let $\text{VaR}_\alpha(X)$ be the value-at-risk of the distribution of X at the risk level α . Show that

$$\lim_{\alpha \rightarrow 0} \text{VaR}_\alpha(X_1 + X_2) > 2\text{VaR}_\alpha(X_1)$$

for $\gamma < 1$, and that the inequality is reversed when $\gamma > 1$. Give an interpretation in terms of subadditivity. Since $\lim_{\alpha \rightarrow 0} \text{VaR}_\alpha(X_1) = \infty$, the previous questions shows that

$$\begin{aligned} P\{X_1 + X_2 > 2\text{VaR}_\alpha(X_1)\} &\sim 2 \frac{1}{2^\gamma \text{VaR}_\alpha(X_1)^\gamma} L\{2\text{VaR}_\alpha(X_1)\} \\ &= 2^{1-\gamma} \alpha \frac{L\{2\text{VaR}_\alpha(X_1)\}}{L\{\text{VaR}_\alpha(X_1)\}} \\ &\sim 2^{1-\gamma} \alpha \end{aligned}$$

as $\alpha \rightarrow 0$. Note that $2^{1-\gamma}\alpha > \alpha$ when $\gamma < 1$, which entails the result. For extreme risk levels, the VaR is thus subadditive for $\gamma > 1$, but overadditive for heavy-tailed distributions such that $\gamma < 1$, which is arguably a good property.